

MT 503: Semiconductor Materials and Devices

Course Code	MT 503
Course Name	Semiconductor Materials and Devices
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech. (Elective) /M.Tech./Ph.D.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- Introduction:** History of semiconductors, semiconductor industry & market. Comprehensive understanding of semiconductor materials and their fundamental properties. Crystal properties, Intrinsic and extrinsic semiconductors, n-type and p-type semiconductors. Carrier dynamics, doping, and semiconductor junctions. Introduction to heavily doped semiconductors and their applications in modern electronics and devices. [8 hours]
- Semiconductor Crystals and Band Structure:** Semiconductor crystal properties (amorphous, crystalline, and polycrystalline), atoms, electrons, holes, and Schrodinger equation. Energy band formation, density of states, and fermi distributions in semiconductors, direct and indirect band gap semiconductors. Energy band diagram of semiconductor junctions, p-n junction, built-in voltage, electric field and potential distributions in p-n junctions and heterojunctions. Concept of holes, and carrier concentration. [6 hours]
- Carrier in Semiconductors:** Effective mass: heavy and light mass carriers, carrier concentration, excess carriers, charge neutrality, carrier transport (diffusion & drift) in semiconductors. Basics of conductivity, mobility, and resistivity in semiconductors, doping in semiconductors, and heavily doped semiconductors. Description of mobility and generation-recombination effects. Introduction to Excitons and optical properties, Phonons: formation and properties, lattice structure and vibrations, optical and acoustic phonons, thermal conductivity in semiconductors. [10 hours]
- Semiconductor Contacts and Devices:** Metal-semiconductor contacts, Schottky barriers, and ohmic contacts, field effect transistors (FET) metal-oxide-semiconductor (MOS) capacitor, metal-oxide-semiconductor field effect transistors (MOSFETs), MOSFET I-V equation, mobility calculation, short/long channel MOSFET. DRAM & RRAM, PN diodes: principles and operation, thin film transistors (TFT), active matrix organic LED (AMOLED). Electrical characterization in various semiconductor devices. Optoelectronic Devices; solar cell, photodetectors, light emitting diodes (LED) and semiconductor lasers. [10 hours]
- Advanced Micro/Nano Fabrication Technology:** Basic micro/nanofabrication processes. Crystal growth and wafer preparation, Methods for p-n junction formation, Thin-film deposition processes. Doping techniques. Heavily doped semiconductors, impact of doping on electronic properties, application of semiconductors in modern electronics. Photolithography and etching. Advanced micro/nanofabrication technologies such as EUV, thin film transistors (TFT), complementary metal oxide semiconductors (CMOS), and 2D material integration. Basic principles of low-dimensional nanodevices. [8 hours]

Laboratory Modules

1. NA

Text Books

1. S. M. Sze, **Physics of Semiconductor Devices**, 4th Edition, John Wiley & Sons, Inc, 2021
2. C. Kittel, **Introduction to Solid State Physics**, 8th Edition, John Wiley & Sons, 2005.
3. N. W. Ashcroft and N. D. Mermin, **Solid State Physics**, Brooks Cole, 1976.
4. Thin Film Materials: Stress, **Defect Formation and Surface Evolution**.
5. Dieter K. Schroder, **Semiconductor Material and Device Characterization**, 3rd Edition, Wiley, 2015

Any other Study Material

1. Students will be supplied with published journal articles as reference materials to support their studies.